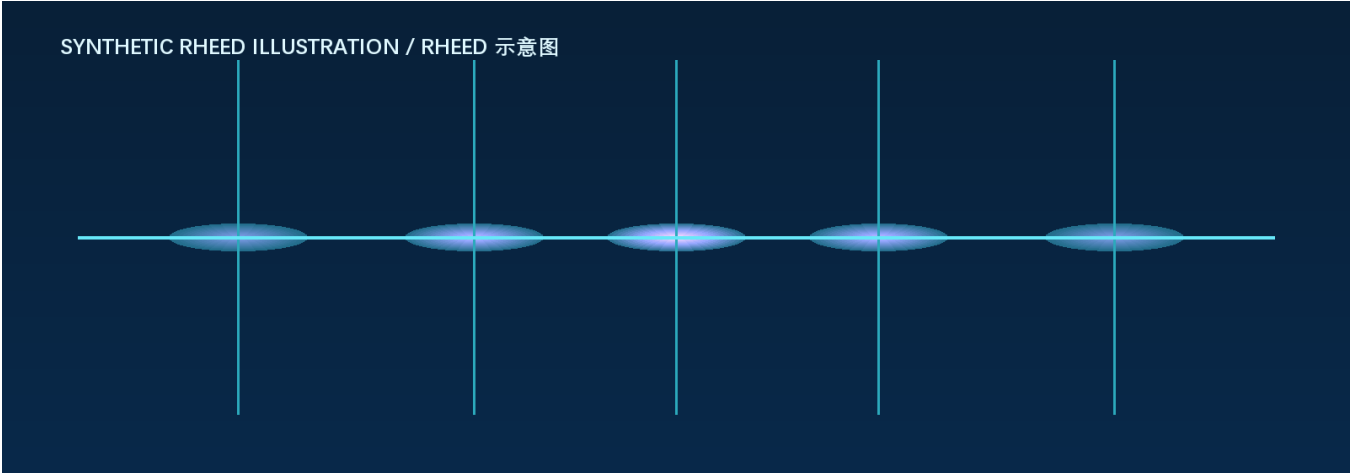


Ch-MBE GUI 与 RHEED 后处理标注

Ch-MBE GUI and RHEED Post-processing Labeling



核心边界 / Core boundary
后处理标注器只读取归档数据，不连接或控制相机、温度计、电源及其他仪器。
The post-processing labeler reads archived data only. It does not connect to or control cameras, pyrometers, power supplies, or other instruments.

快速入口 / Quick entry	双击 / Double-click
Ch-MBE 实时 GUI / Live GUI	Start Ch-MBE Growth Monitor.cmd
RHEED 离线标注器 / Offline labeler	Start RHEED Post-processing Labeler.cmd
安装或更新桌面快捷方式 / Install or refresh shortcuts	Install AI4MBE Desktop Shortcuts.cmd

版本：Manual v1.0，发布日期：2026-08-10。适用于 Windows 上的 Ch-MBE 工作站。
Version: Manual v1.0, issued 2026-08-10. Intended for the Windows Ch-MBE workstation.

1 先理解两条独立路径

Understand the two independent paths

实时 GUI 用于采集和显示；离线标注器只处理已经结束并打包的会话。不要把离线工具的结果解释为实时控制指令。

The live GUI performs acquisition and display. The offline labeler works only on completed archived sessions. Never interpret offline output as a real-time control instruction.

数据流 / Data flow



Offline labeler never controls instruments. / 离线标注器不控制仪器。

本手册包含 / This manual covers

页 / Page	内容 / Topic	结果 / Outcome
3	快捷方式和 Ch-MBE GUI 启动 / Shortcuts and live GUI startup	确认启动正确且正常退出
4	标注器桌面界面 / Labeler desktop UI	选择输入、构建、打开、验证
5-6	输入和安全构建 / Inputs and safe build	得到完整本地报告目录
7-8	分段标注和显示控制 / Segments and display controls	按真实保存帧建立可追溯标签
9	导出和验证 / Export and validation	得到 canonical JSON 并 fail-closed 验证
10-12	数据安全、排障和版本 / Safety, troubleshooting, version	安全交接和复现

术语 / Terminology

RHEED 重构标签、采集质量 QC、FeSe 薄膜质量是三个不同概念。

Reconstruction labels, acquisition-quality QC, and FeSe film quality are three different concepts.

2 安装快捷方式并启动 Ch-MBE GUI

Install shortcuts and start the Ch-MBE GUI

首次设置或版本更新后 / First setup or after an update

- 1 在 GUI 仓库根目录双击 Install AI4MBE Desktop Shortcuts.cmd。
From the GUI repository root, double-click Install AI4MBE Desktop Shortcuts.cmd.
- 2 确认桌面上出现或更新两个入口：Ch-MBE Growth Monitor 和 RHEED Post-processing Labeler。
Confirm that the desktop now has refreshed entries for Ch-MBE Growth Monitor and RHEED Post-processing Labeler.
- 3 若 Windows 显示意外的安全警告，不要绕过；停止并联系维护者核对文件来源和 commit。
If Windows shows an unexpected security warning, do not bypass it. Stop and ask the maintainer to verify the file source and commit.

每次实验启动 / Start each experiment

- 1 双击 Start Ch-MBE Growth Monitor.cmd 或对应桌面快捷方式。不要从 O-MBE 入口启动。
Double-click Start Ch-MBE Growth Monitor.cmd or its desktop shortcut. Do not use the O-MBE launcher.
- 2 等待主窗口出现，确认标题或 chamber 信息为 Ch-MBE，并查看各设备状态。
Wait for the main window, confirm the title or chamber identity is Ch-MBE, and inspect each device status.
- 3 只有在操作员准备好后才按实验 SOP ARM 或开始会话；双击启动本身不授权改变仪器设定值。
ARM or begin a session only when the operator is ready under the experiment SOP. Double-click startup alone does not authorize setpoint changes.
- 4 结束时使用 GUI 的正常关闭方式并等待窗口消失。不要在日志写入期间强制结束 Python。
Close through the GUI normally and wait for the window to disappear. Do not force-kill Python while logs are being written.

预期环境 / Expected environment

启动器日志位于 %LOCALAPPDATA%\AI4MBE\LauncherLogs，并显示实际解析到的 Python 和仓库路径。核对其符合该工作站配置；不要假设固定盘符，也不要随意新增全局环境变量。

Launcher logs are under %LOCALAPPDATA%\AI4MBE\LauncherLogs and show the resolved Python and repository paths. Verify them against that workstation's configuration; do not assume a fixed drive letter or add global environment variables ad hoc.

启动后快速检查 / Quick check after startup

检查 / Check	正常 / Expected	异常时 / If abnormal
Chamber	显示 Ch-MBE 配置	立即关闭，核对启动器
RHEED	新帧/sequence 继续更新	不要保存旧帧；查窗口/相机状态
Temperature	读数与 valid/connected 一致	记录模式和错误；不要猜测串口参数
Logs	会话开始后写入指定目录	先保留错误文本，再联系维护者

3 打开 RHEED 后处理标注器

Open the RHEED post-processing labeler

- 1

双击 Start RHEED Post-processing Labeler.cmd 或其桌面快捷方式。
Double-click Start RHEED Post-processing Labeler.cmd or its desktop shortcut.
- 2

标注器应显示 Build、Open existing report、Validate 和 Open PDF manual。它不会启动 Growth Monitor。
The labeler should show Build, Open existing report, Validate, and Open PDF manual. It does not start Growth Monitor.

RHEED Post-processing and Temporal Labeling

离线构建时间轴并标记重构分段 / Build an offline timeline and label temporal segments

模型输出可见：仅为 model-assisted review，不是 blind-gold。
Model outputs are visible: exports are not eligible as blind-gold labels.

Session ZIP / 会话压缩包

<local-data-root>\growth_session.zip

Browse

Ordered model pairs
有序模型配对

Prediction CSV	Model-spec JSON
<local-data-root>\predictions.csv	<local-data-root>\model_spec.json

Add pair

Remove selected

Move up

Move down

Output directory / 输出目录

<local-data-root>\growth_session_labeling_report

Browse

Report title / 报告标题

RHEED reconstruction timeline

WebP quality

78

Build report and open

Status / 状态: Ready

Messages appear here / 消息显示在此

SYNTHETIC UI / 界面示意

区域 / Area	用途 / Purpose
Session ZIP	选择一个归档的 Growth Monitor 会话 / Select one archived session
Model inputs	每行配对一个 predictions CSV 和 model-spec JSON / One positional pair per row
Output directory	必须是新目录或空目录，推荐仓库外 / New or empty, preferably outside Git
Message log	保留构建/验证成功或失败的完整信息 / Preserve complete build or validation output

离线 / Offline

此窗口可以在没有仪器软件的电脑上使用，只要输入文件完整且 Python 环境正确。
This window can run on a computer without instrument software, provided the archived inputs and Python environment are complete.

4 准备输入并核对溯源

Prepare inputs and verify provenance

会话压缩包 / Session ZIP

压缩包必须包含一个 session_metadata.json、一个 heartbeat_log.csv，以及 heartbeat 记录引用的 frames/ 图像。不要解压后手动重命名帧。

The archive must contain one session_metadata.json, one heartbeat_log.csv, and the frames/ images referenced by heartbeat rows. Do not manually rename extracted frames.

```
growth_session.zip
  session_metadata.json
  heartbeat_log.csv
  frames/
    heartbeat_000001_...bmp
    heartbeat_000002_...bmp
```

预测表与模型说明必须成对 / Predictions and model specs are positional pairs

文件 / File	必须包含 / Must contain	检查 / Check
Prediction CSV	frame_index, heartbeat_idx, elapsed_s, captured_at_utc, capture_sequence, frame_name, frame_sha256	行顺序与保存帧完全一致
Model-spec JSON	key, title, classes, probability_columns, status, provenance	类别顺序与概率列一一对应
Session ZIP	metadata, heartbeat, frames	每个 frame_path 在 ZIP 内存在

prediction CSV 的 frame_index 可以从 0 或 1 开始，但随后必须连续。heartbeat_idx、capture_sequence 和 elapsed_s 可以有间隔；工具按真实保存帧和真实时间工作，不假设严格 1 Hz。

Prediction frame_index may begin at 0 or 1, then must remain contiguous. heartbeat_idx, capture_sequence, and elapsed_s may have gaps. The tool follows actual saved frames and recorded times; it never assumes exact 1 Hz sampling.

Fail closed

任何行数、顺序、时间、sequence、文件名或 SHA-256 不匹配都会停止构建。不要编辑输入来绕过错误。

Any mismatch in row count, order, time, sequence, filename, or SHA-256 stops the build. Do not edit inputs to bypass the error.

推荐目录 / Recommended local layout

```
<local-data-root>\2026-08-10_anneal\
  source\growth_session.zip
  predictions\model_A.csv
  specs\model_A.json
  output\                # choose this new/empty folder
  exports\               # annotation JSON/CSV
```

5 安全构建并打开报告

Build and open a report safely

- 1 在 Session ZIP 中选择原始归档副本。
Choose the preserved archive copy in Session ZIP.
- 2 对每个要展示的模型添加一行 Prediction CSV + Model-spec JSON；顺序必须配对。
Add one Prediction CSV + Model-spec JSON row for every model to display; pairing is positional.
- 3 选择仓库外的新目录或空目录。桌面标注器不会覆盖非空目录。
Choose a new or empty directory outside the repository. The desktop labeler will not overwrite a non-empty directory.
- 4 填写报告标题；review quality 只影响审阅图，不改变原始图或模型输出。默认 78 适合一般使用。
Enter a report title. Review quality changes only review images, never archived pixels or predictions. The default 78 is suitable for routine use.
- 5 点击 Build report and open，等待状态为成功并自动打开 interactive_report.html。
Click Build report and open. Wait for a success status and automatic opening of interactive_report.html.

PowerShell 备用方式 / PowerShell fallback

```
Set-Location '<GUI repository>'
conda activate ai4mbe-gui
python -m tools.rheed_postprocessing_labeling build `
  --session '<local-data-root>\growth_session.zip' `
  --predictions '<local-data-root>\predictions.csv' `
  --model-spec '<local-data-root>\model_spec.json' `
  --output-dir '<local-data-root>\labeling-output'
```

不要只复制 HTML / Do not copy the HTML alone

报告目录包含 interactive_report.html、images/、vendor/ 和 run_manifest.json。移动或交接时复制整个目录；同时在报告目录外单独保留原始 prediction CSV 和 model-spec JSON。

The report directory contains interactive_report.html, images/, vendor/, and run_manifest.json. Copy the whole directory when moving or handing it off, and preserve the original prediction CSV and model-spec JSON separately outside the report directory.

--overwrite 的边界 / --overwrite boundary

桌面界面始终拒绝非空输出目录。高级 CLI 的 --overwrite 只允许替换已含 run_manifest.json 和 interactive_report.html 的有效工具输出，并在替换前先验证本次输入；它仍拒绝文件系统根、仓库/源码祖先、当前目录和含输入文件的目录。

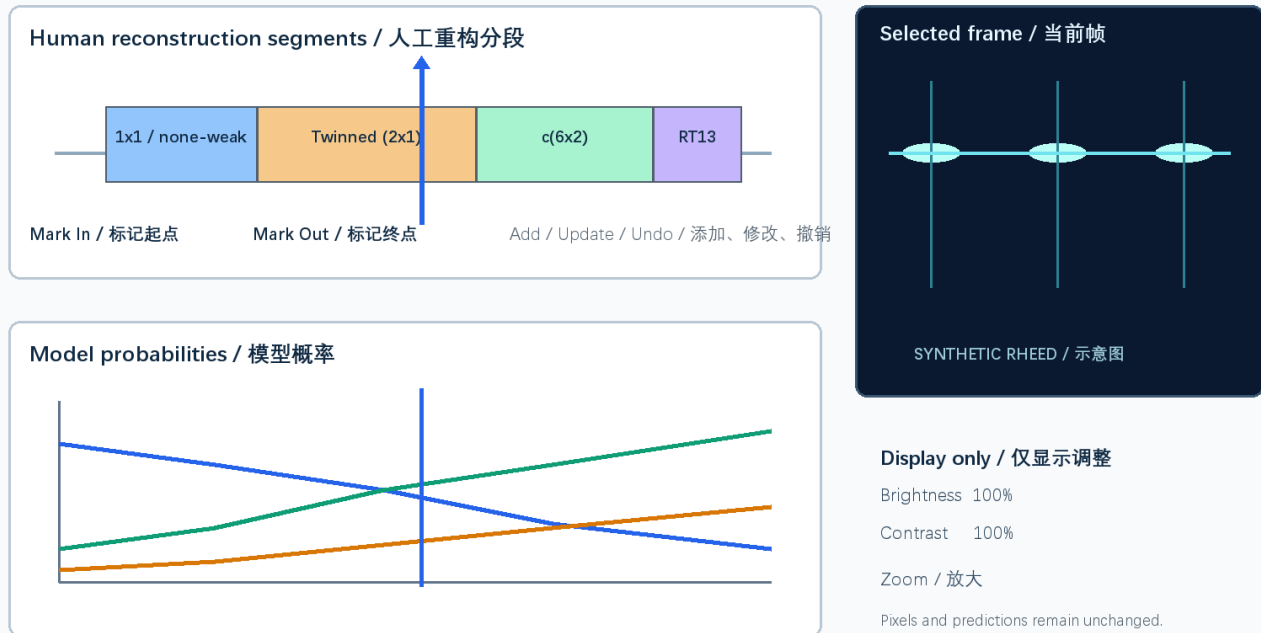
The desktop UI always refuses a non-empty output directory. Advanced CLI --overwrite may replace only a recognized tool output containing run_manifest.json and interactive_report.html, and it validates the new inputs before replacement. It still refuses filesystem roots, repository/source ancestors, the current directory, and directories containing inputs.

6 像剪辑软件一样做时间分段标注

Label temporal segments like an editing timeline

RHEED reconstruction timeline / RHEED 重构时间轴

Model-assisted review - not blind-gold / 模型辅助审阅 - 非盲标金标准



1

拖动 Frame playhead 或点击任意模型曲线，找到分段的第一张保存帧。点击 Mark In。
Move the Frame playhead or click a model plot to find the first saved frame. Click Mark In.

2

移动到该分段的最后一张保存帧，点击 Mark Out。Out 是包含端点。
Move to the final saved frame in the segment and click Mark Out. Out is inclusive.

3

选择重构标签，填写 Labeler，按需添加 Notes，然后点击 Add segment。
Choose a reconstruction label, enter Labeler, add Notes if useful, then click Add segment.

4

点击已保存分段可修改；Update 保留 annotation_id。相邻分段允许，重叠分段会被拒绝。
Select a saved segment to edit it. Update preserves annotation_id. Adjacent segments are allowed; overlaps are rejected.

界面显示和导出的 saved-frame ordinal 从 1 开始。每个端点同时记录 heartbeat、capture sequence、UTC 和 frame SHA-256，因此即使采样有间隔也可追溯。

Visible and exported saved-frame ordinals are 1-based. Every endpoint also records heartbeat, capture sequence, UTC, and frame SHA-256, so it remains traceable despite sampling gaps.

7 审阅、显示调整和草稿

Review, display controls, and drafts

所有时间指示器保持同步 / All time indicators stay synchronized

操作 / Action	应同时变化 / Must update together	不应变化 / Must not change
拖动主时间轴	当前 RHEED 图、模型 guide、放大窗口时间轴	归档文件和预测值
点击模型曲线	主 playhead、图像、metadata	已保存分段
Brightness / Contrast	屏幕显示	像素、SHA-256、模型输出
Enlarge	放大视图和可调时间轴	原图、标注边界

草稿与编辑 / Draft and editing

- 浏览器按 dataset_id 在 localStorage 中保存草稿；它不是长期备份。
The browser stores a dataset-scoped draft in localStorage; this is not a durable backup.
- 频繁 Export JSON，特别是在长会话、换电脑、换浏览器或清理缓存前。
Export JSON frequently, especially for long sessions and before changing computers, browsers, or cache settings.
- Delete selected 需要确认；Undo 只恢复最近一次标注修改。
Delete selected requires confirmation. Undo restores only the most recent annotation mutation.
- Import JSON 会先严格验证；失败时当前分段集合保持不变。
Import JSON validates before replacement. On failure, the current segment set remains unchanged.

标签解释 / Interpretation

标签描述审阅者在该时间区间看到的主要表面重构类别。unknown 表示无法可靠判断。不要用模型 argmax 自动覆盖人的选择。
Labels describe the dominant surface-reconstruction category seen by the reviewer over that interval. Use unknown when a reliable decision is not possible. Never overwrite the human choice with model argmax.

显示增强不是数据增强 / Display adjustment is not data modification
亮度、对比度和放大只帮助观察。导出会记录显示设置，但不会改写归档图像。
Brightness, contrast, and zoom aid inspection only. Exports record display settings but do not rewrite archived images.

建议审阅节奏 / Recommended review rhythm

阶段 / Stage	操作 / Action
粗看 / Survey	快速拖动全程，识别明显转变和数据缺口
精标 / Boundary pass	逐帧确认 In/Out，并看 metadata
复查 / Review	检查未覆盖区、重叠错误、unknown 和备注
保存 / Save	导出 JSON，随后运行验证

8 导出、导入与 fail-closed 验证

Export, import, and fail-closed validation

JSON 是 canonical 文件 / JSON is the canonical artifact

Export JSON 用于继续标注、验证和下游处理；Export CSV 便于会议和表格分析，但不能替代 JSON 的完整嵌套溯源。
Use Export JSON for resume, validation, and downstream processing. Export CSV is convenient for meetings and tabular analysis, but it does not replace the complete nested provenance in JSON.

绑定字段 / Binding	作用 / Purpose
dataset_id + source_archive_sha256	绑定原始会话 / Bind the source session
ordered_frame_fingerprint	绑定帧顺序与逐帧 SHA / Bind order and per-frame SHA
model_review_context_fingerprint	绑定审阅时可见的 predictions/specs / Bind visible model context
endpoint provenance	绑定 frame ordinal、heartbeat、UTC、sequence 和 SHA / Bind each boundary
model_outputs_visible=true	披露模型可见 / Disclose model visibility
eligible_for_gold=false	禁止误当 blind-gold / Prevent blind-gold misuse

在桌面标注器验证 / Validate in the desktop labeler

- 1

选择原始 interactive_report.html。
Select the original interactive_report.html.
- 2

选择刚导出的 annotation JSON。
Select the exported annotation JSON.
- 3

点击 Validate annotations；只有显示 validation passed 才进入下一步。
Click Validate annotations. Continue only after validation passes.

PowerShell 验证 / PowerShell validation

```
python -m tools.rheed_postprocessing_labeling validate `
--report '<local-data-root>\labeling-output\interactive_report.html' `
--annotations '<local-data-root>\exports\segment_annotations.json'
```

验证失败不要绕过 / Never bypass a failure
wrong run、模型上下文变化、端点被改、重叠或 gold 声明都会失败。回到正确的报告和原始导出重新核对。
A wrong run, changed model context, altered endpoint, overlap, or gold claim all fail validation. Return to the correct report and original export and investigate.

9 数据科学与科学边界

Data safety and scientific boundaries

仓库中允许和禁止的内容 / What belongs in Git

可进入 Git / May be tracked	不得默认进入 Git / Keep out by default
工具代码、模板、测试、说明文档	真实 session ZIP、原始 RHEED 图、sensor logs
model_spec.example.json	真实 predictions、生成报告、标注 JSON/CSV
合成测试图和本手册示意图	checkpoint、未公开结果、浏览器截图

安全处理顺序 / Safe handling sequence

- 1

把原始 session ZIP 设为只读或保留不可修改副本，并记录 SHA-256。

Keep the source session ZIP read-only or preserve an immutable copy, and record its SHA-256.
- 2

在仓库外创建专用工作目录；报告、导出和截图都留在这里。

Create a dedicated working directory outside the repository. Keep reports, exports, and screenshots there.
- 3

交接时复制完整报告目录和 canonical JSON，并在外部单独保留原始 prediction CSV、model-spec JSON 与 SHA 清单；不要只发 HTML。

For handoff, copy the whole report directory and canonical JSON, and separately preserve the original prediction CSV, model-spec JSON, and SHA manifest; never send only the HTML.
- 4

验证通过后再进行统计或训练数据整理，并保留原始导出不变。

Run statistics or training-data preparation only after validation, and preserve the original export unchanged.
- 不是 blind-gold / Not blind-gold

标注时模型曲线和 lossy review image 可见，所以这些标签只能作为 model-assisted review。正式 blind-gold 必须使用隐藏模型输出的独立流程。

Model plots and lossy review images are visible during labeling, so these labels are model-assisted review only. Formal blind-gold data requires a separate workflow that hides model outputs.
- 不要扩大结论 / Do not over-interpret

当前分类目标是裸 STO 表面重构。不要把结果直接解释为 FeSe 薄膜质量，也不要把 reconstruction unknown 等同于 QC reject。

The current classification target is bare STO surface reconstruction. Do not directly interpret it as FeSe film quality, and do not equate reconstruction unknown with QC reject.

10 故障排查

Troubleshooting

先保存错误文本、当前 branch/commit 和所选文件路径。不要用删除环境、全局变量或强制 Git 重置作为第一反应。若出现 AI4MBE live-driver warning，请记录缺失模块清单和启动器日志；dummy 及不受影响的模式仍可使用，但修复环境前不得用缺失驱动 ARM 生产会话。这是预期的可见告警，不应临时安装依赖来绕过。

First preserve the exact error text, branch/commit, and selected paths. Do not begin by deleting environments, changing global variables, or force-resetting Git. If AI4MBE live-driver warning appears, record its missing-module list and launcher log. Dummy and unaffected modes remain available, but do not ARM a production session that needs a missing driver until the environment is repaired. This is expected fail-visible behavior, not a prompt for ad-hoc installation.

症状 / Symptom	可能原因 / Likely cause	安全操作 / Safe action
桌面快捷方式缺失或指向旧目录	安装脚本未运行或 checkout 已移动	从当前仓库根目录运行 Install AI4MBE Desktop Shortcuts.cmd
Growth Monitor 启动后 chamber 不对	用了错误入口	正常关闭；只用 Start Ch-MBE Growth Monitor.cmd
启动窗口立即消失	环境或依赖错误	查看 %LOCALAPPDATA%\AI4MBE\LauncherLogs 最新日志；必要时从 PowerShell 运行同一 .cmd
温度/RHEED 无读数	仪器接口、vendor 窗口或模式问题	记录 connected/error/mode；不要随意改全局环境变量
标注器拒绝 Build	输入缺失、pair 不匹配或输出目录非空	逐项核对 Session、每行 pair 和新输出目录
构建提示 provenance mismatch	预测不属于该 ZIP 或行被改	找回对应预测与 spec；不要编辑 CSV 绕过
HTML 无图或曲线	只复制了 HTML 或本地 asset 缺失	恢复完整 report 目录；保持相对路径
草稿消失	换浏览器/电脑或清理 localStorage	导入最近一次 JSON；以后更频繁导出
Import/Validate 失败	run/context/endpoint/overlap 不一致	使用原始报告和原始 JSON，查看具体 fail-closed 信息

需要提供给维护者的最小证据 / Minimum evidence for the maintainer

- 错误信息全文和发生时间；不要只发“不能用”。
Full error text and occurrence time; do not send only 'it does not work'.
- git status --short --branch 与 git rev-parse HEAD。
git status --short --branch and git rev-parse HEAD.
- 启动器名称、环境路径、输入文件名和输出目录；敏感数据不要发到公共频道。
Launcher name, environment path, input filenames, and output directory; keep sensitive data out of public channels.
- 标注问题附 report manifest SHA 和 annotation JSON SHA，不需要先发送原始图。
For labeling issues, provide report-manifest and annotation-JSON SHA values; raw images are not initially required.

11 版本核对与一页检查表

Version verification and one-page checklist

实验或标注前记录版本 / Record versions before acquisition or labeling

```
Set-Location '<GUI repository>'
git status --short --branch
git rev-parse HEAD
conda env list
conda activate ai4mbe-gui
python --version
python -m tools.rheed_postprocessing_labeling --help
```

GUI 的实际 Python 和仓库路径以启动器日志为准；AI4MBE_GUI_PYTHON 仅在该工作站已配置时读取，不要临时创建。若 branch/commit 与团队指定版本不同，或 git status 显示未知改动，先停止并核对。

Use the resolved Python and repository paths printed by the launcher log. Read AI4MBE_GUI_PYTHON only when it is already configured on that workstation; do not create it ad hoc. If branch/commit differs from the team-specified version, or git status shows unknown changes, stop and verify.

完整性哈希 / Integrity hashes

```
Get-FileHash '<local-data-root>\growth_session.zip' -Algorithm SHA256
Get-FileHash '<local-data-root>\exports\segment_annotations.json' -Algorithm SHA256
Get-FileHash '.\docs\RHEED_GUI_Postprocessing_Labeling_User_Manual.pdf' -Algorithm SHA256
```

快速检查表 / Quick checklist

☐	操作员确认 / Operator confirmation
☐	使用 Start Ch-MBE Growth Monitor.cmd，chamber 显示正确
☐	实时状态、帧 sequence、温度和日志行为符合实验 SOP
☐	原始 session ZIP 有只读副本和 SHA-256
☐	用 Start RHEED Post-processing Labeler.cmd 打开离线工具
☐	每个 prediction CSV 与 model-spec JSON 正确配对
☐	输出目录在 Git 外且为新目录或空目录
☐	时间分段无重叠，边界逐帧核对，unknown/notes 使用合理
☐	Export JSON 后使用原始 interactive_report.html 验证通过
☐	记录 branch、commit、环境、报告 manifest 和导出 SHA
☐	交接完整 report 目录；声明 model-assisted、not blind-gold

停止条件 / Stop condition

出现无法解释的仪器状态、数据旧值、版本不明、provenance mismatch 或验证失败时，停止继续采集/标注并联系维护者。

Stop and contact the maintainer if instrument state, stale data, version identity, provenance, or validation cannot be explained.

本手册中的界面和 RHEED 图均为合成示意，不包含真实实验数据。

All interface and RHEED images in this manual are synthetic illustrations and contain no experimental data.